

Dev [Boost]

GitHub Copilot

GitHub Copilot Intermediate Trainings

Deep dive into Chat Participants,
Variables, Context and specific use
cases with GitHub Copilot.



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Introductions

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Accepting Terms of Use

For you to begin using GitHub Copilot, it is critical to accept the Siemens Terms of Use relating to its usage.

T - Tool: Controlled by developer and developer decides on if and in what form the tool is integrated.

E - Evaluate Proposed Code: Treat the output as a draft and ensure it's reviewed by a human developer.

A - Adapt or Edit (whenever meaningful or required): Significant changes made by the developer will render the code protected as independent work)

M - Manage Use: Developer determines how, where, and in what form the generated code is used.

GitHub Copilot for Business Offered by github.siemens.cloud

Hallo, Thomas. Wenn Sie dieses Formular absenden, sieht der Eigentümer Ihren Namen und Ihre E-Mail-Adresse.

* Erforderlich

Participation request

By clicking on the send button you agree to the information below

1. General information:

This is a service offered by github.siemens.cloud - your internal ACP-3 managed GitHub Enterprise provider!

We are using GitHub Copilot for Business, therefore GitHub Copilot does not store your code and does not learn from your code!

GitHub Copilot Wiki: <https://pages.github.siemens.cloud/ghes-gps/github-enterprise-wiki/documentation/other/copilot/>

Siemens Viva

Engage: <https://engage.cloud.microsoft/main/groups/eyJfdHlwZSI6IkdvdjZVwliwWQ0iixODY1MDg5OTI1MTI0>

Unit approval status:

- Approved: DI, ADV (ADV AI policy applies), SFS, Service & Governance, RC, SI (except SI BSW)
- Waiting for approval: SMO

Costs:

The costs of GitHub Copilot for Business are 13€ per user per month and is charged to the Cost Center of the Department

How can I use GitHub Copilot for Business after I finished this self-service?

After you pressed the send button, you'll get an E-Mail with an exact description.

Terms of Use:

- The approval and distribution of the Copilot licenses is only done by the GitHub Enterprise team via this self-service, don't buy separate subscriptions and/or interact with the GitHub Copilot sales team by yourself!
 - Do not use the subscriptions of GitHub Copilot offered for private users for work purposes!
- Your use of the service is subject to the compliance with the Acceptable Use Policies of GitHub Copilot as published here: <https://docs.github.com/en/site-policy/acceptable-use-policies>. For example, you may not use GitHub Copilot to generate suggestions whose use you know (or reasonably should know) would be unlawful or would infringe on the rights of others.
- Mark generated code: You must mark the suggestions generated by GitHub Copilot as defined here: <https://pages.github.siemens.cloud/ghes-gps/github-enterprise-wiki/documentation/other/copilot/#documentation-of-github-copilot-usage>.
- Security: GitHub Copilot is approved for all ACP levels, but the ACP level of where you host your source code is relevant. Follow the mandatory SFeRA Secure Software Development Standard and Product and Solution Security requirements.
 - Do not blindly trust the generated code, validate and quality check if it's suitable for the intended purpose and following secure coding practices. We recommend to also check the code quality with tools like SonarQube and to have a high test coverage for the Copilot generated code
 - Train the usage of GitHub Copilot and check the Tips & Tricks (see Wiki).
- Export Control:
 - If the export control classification of your product is different than AL=N & ECCN=N or EAR99 then the upload of controlled technology is not allowed on third party servers because of potential license restrictions. Please contact your EX responsible (<https://intranet.legal-compliance.siemens.com/compliance/ico/#icoContacts>) to clarify the usage of GitHub Copilot.
 - Currently, the usage of GitHub Copilot is not allowed in embargo countries (Belarus, Cuba, Iran, North Korea, Russia, Syria)
- These terms are subject to change and will be adapted as necessary, e.g. due to additional legal requirements.

Advanta only:

- I confirm that I successfully passed the mandatory AI training: https://mylearningworld.siemens.com/web/en/app/toc/lex_auth_013995406482554880583/overview

✕

☒ I accept the Terms of Use

Absenden

Microsoft 365

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Chat Participants

Deeper dive into @workspace,
@terminal and @vscode agents



GitHub Chat Participants & Their Slash Commands:

@workspace agent

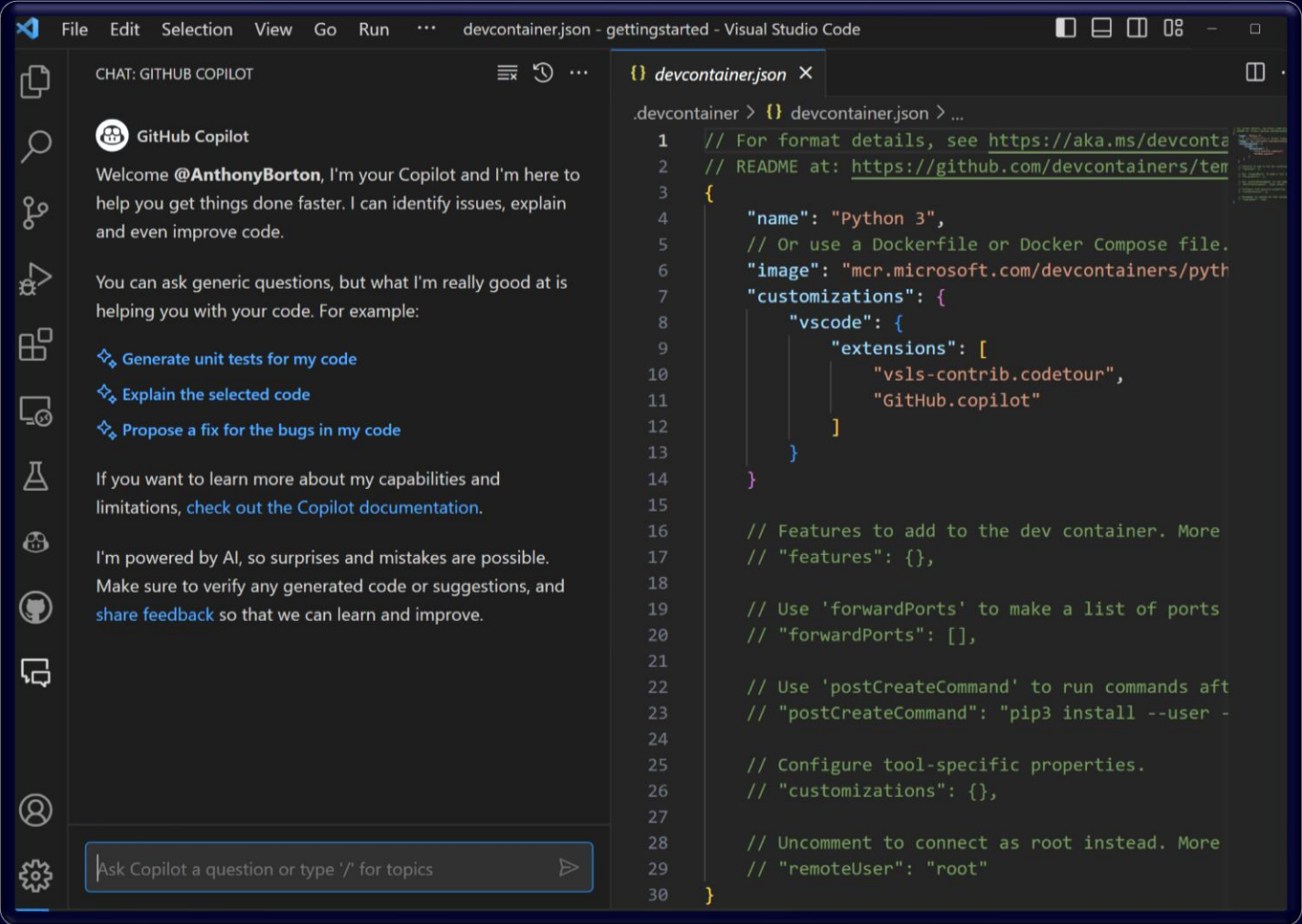
- /explain** Explains how the selected code works
- /fix** Proposes a fix for the problems in selected code
- /new** Create a new project based on a natural language description
- /newNotebook** Create a new Jupyter Notebook based on your description
- /tests** Generate unit tests for the selected code

@vscode agent

- /runCommand** Search for and execute a VS Code command
- /search** Generate search queries to find code

@terminal agent

- /explain** Explain terminal errors and commands



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Enriching Chat Context

How to focus Copilot's attention on specific parts of your code base.





Adding to a chat context

GitHub Copilot Chat's context is built up from:

- Active chat thread
- Your new prompt
- Files/parts of files in your workspace that have been added to Chat context

Different ways of adding code to the current context:

- Variables: **#editor**
- Variables: **#file**
- Variables: **#selection**
- Variables: **#terminalLastCommand**
- Variables: **#terminalSelection**
- Variables: **#vscodeApi**
- Context Menu: **Add File**
- Context Menu: **Add Selection**
- Input box: **Attach Icon**

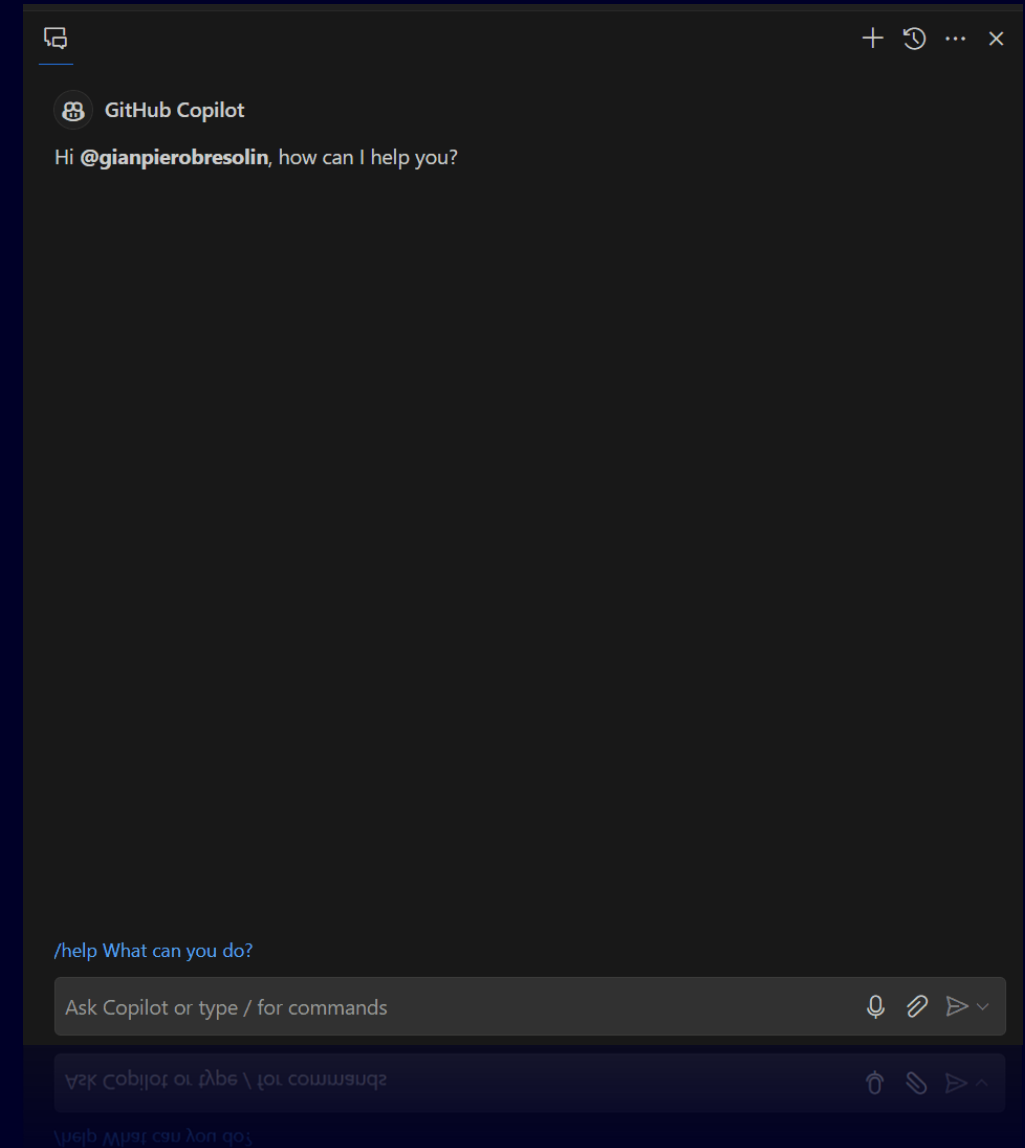
Referenced code can be seen in chat under **references**.

Chat Management:

Remember that each chat represents a context that is sent to GitHub when you submit a new prompt. This context is used when generating the next response, which means **you can improve your responses by keeping each chat on a specific topic.**

This context also has a limited size, which is why it's important to remember the following:

- ✓ Start a new chat when discussing a new topic using the **+** icon
- ✓ Revisit old chats using the **Show Chat** history button
- ✓ Remove chats that are no longer relevant using the **x** in the top right corner of the chat
- ✓ Remove irrelevant prompts from the chat if the response isn't what you were looking for using the **x** icon next to the prompt

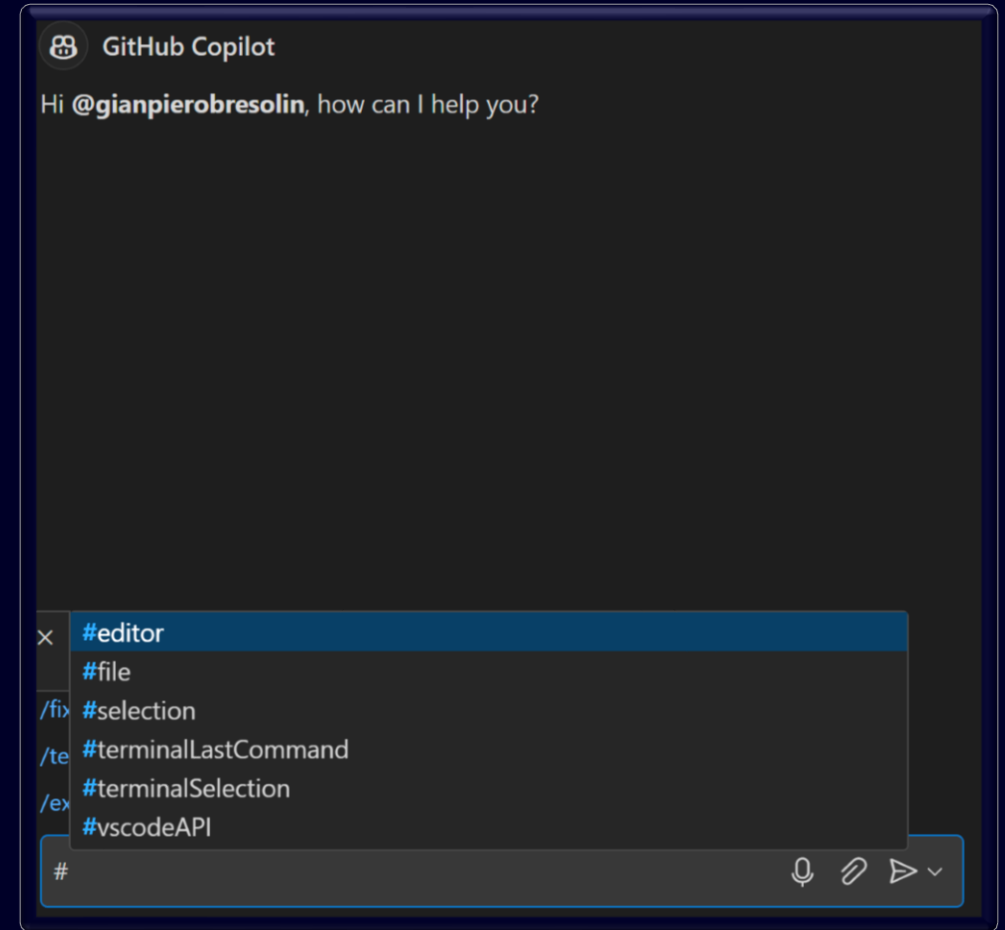


Using Variables & Attach Context to enrich context:

Variables allow you to focus GitHub Copilot on a specific area within VS Code which is important when working with larger code bases or when working with the terminal.

Variables can be added to the context using a # variable or by clicking the paperclip (**Attach Context**) icon in the input field.

- ✓ #editor
- ✓ #file
- ✓ #selection
- ✓ #terminalLastCommand
- ✓ #terminalSelection
- ✓ #vscodeAPI



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Specific Use Cases

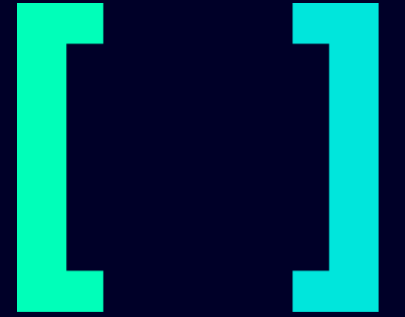
Let's go through some examples of some specific ways you can use GitHub Copilot



Demo

Specific Use Cases





Adapting your code base to GitHub Copilot

Inner sourcing and best practices that you can adopt to improve GitHub Copilot's integration with your code base

What is inner sourcing?

Inner sourcing is an approach modelled on lessons learnt in the open-source community and describes how companies can use open-source project practices to improve the way in which we build software within large organizations.

While effective for humans, teams and organizations, **the same techniques are applicable to getting the best value from GitHub Copilot**, as they involve documenting and sharing code, approaches and techniques in common areas in markdown files that are discoverable by humans *and* generative AI tools like GitHub Copilot.

Adapting your solution for GitHub Copilot

1. Documentation should be in the repository explaining how the application works, design patterns followed, code owners, contribution guidelines, linting rules, style guides etc.
2. Keep guidelines in a separate files based on intention.
3. Follow open-source naming conventions for files: CONTRIBUTION.md, CODEOWNERS, ReadMe.md.
4. Use GitHub Copilot to document code and summarize approaches in markdown files.
5. Reference examples of code that follows best practices within the documentation.
6. Prompts that are effective for generating new code following specific patterns should be checked in with any additional guidance documented.
7. Get GitHub Copilot to summarize chat thread into prompts that can be used in future chats.
8. Prompts that are generally effective for an organization should be shared in Community of Practices/Centers of Excellence.



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THANK YOU!

Questions & Answers

